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EXTENDED ABSTRACT

Sim2Real Conditional Diffusion for Radar Beam Blockage Correction: A Preliminary Study

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1. Introduction and objectives

Ground–based weather radars provide high temporal and spatial resolution precipitation estimates that are indispensable for hydrological modelling, flood forecasting and climate research. Accurate precipitation observations are critical for understanding climate change and associated hydrological responses across scales. However, quantitative precipitation estimation (QPE) derived from radars suffers from systematic biases; terrain or buildings can partially block the radar beam, reducing the returned power and leading to systematic underestimation of reflectivity and precipitation [McRoberts and Nielsen–Gammon(2017)]. Traditional correction methods rely on geometric calculations, climatology or multi–radar fusion, but these approaches are limited by complexity, computational cost and the need for ancillary data [Tan and Chen(2025)]. Recent data–driven methods such as Generative Adversarial Networks (GANs) have shown that deep learning can recover missing precipitation patterns and improve QPE in mountainous regions [Tan and Chen(2025), Liu et al.(2024)]. In this preliminary study we explore a conditional diffusion model for radar beam blockage correction [Ho et al.(2020)]. Our objective is to extend diffusion modelling to the radar QPE domain by combining supervised training on synthetic beam blockage examples with unsupervised inference on real precipitation mosaics.

2. Methods and data

We propose a diffusion-based framework for correcting beam-blockage artefacts in weather radar precipitation mosaics. Our approach operates on the national Météo-France radar mosaic and leverages the operational blockage map to identify regions that are free of beam-blockage effects. Only these clear sectors are retained for training. To generate supervised data, we simulate beam attenuation by applying synthetic azimuthal masks that reproduce realistic blockage patterns to the precipitation fields of the clear sectors. Since the attenuation is artificially introduced, the exact degradation process is known, providing paired data composed of a corrupted observation and its corresponding clean ground-truth precipitation field. A conditional diffusion model is trained on these paired samples. During training, the clean precipitation mosaics are progressively corrupted with Gaussian noise following a forward diffusion process. At each denoising step, the model is conditioned on the synthetically blocked mosaic and learns to predict the injected noise, thereby reconstructing the clean image. This conditioning

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mechanism encourages the model to learn a mapping between blocked structures and their underlying precipitation patterns, enabling the joint removal of synthetic attenuation and diffusion noise. At inference time, the model is applied to real radar mosaics affected by beam-blockage artefacts, for which no clean reference is available. The reverse diffusion process is initialized from Gaussian noise and conditioned on the observed blocked mosaic at each timestep. Through iterative denoising, the model generates a plausible estimate of the blockage mask, which is then removed from the input mosaic to recover an approximation of the original precipitation field.

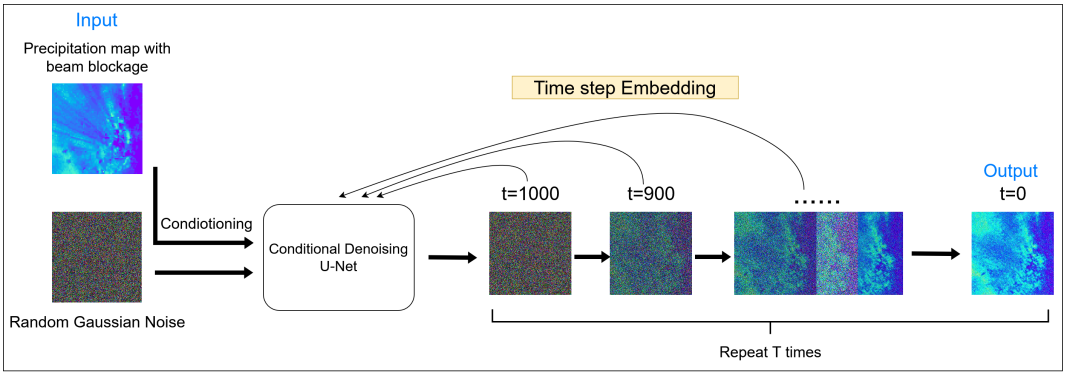

Figure 1. *Overview of the Sim2Real conditional diffusion pipeline during inference. Starting from random noise, the model performs a reverse diffusion process conditioned on the input image with beam blockage to generate a restored image without beam blockage.*

3. Preliminary results and discussion

The accompanying figure 1 illustrates the reverse diffusion sampling procedure during inference. Starting from pure Gaussian noise, the diffusion model performs 1000 denoising steps to progressively generate a beam-blockage-free precipitation map conditioned on the observed corrupted mosaic. Intermediate samples highlight the gradual refinement of the reconstruction and the restoration of spatially coherent precipitation structures. While the model effectively identifies blocked sectors, slight modifications of precipitation values may occur in uncorrupted regions. Future work will target enhanced synthetic mask generation during training to preserve rainfall intensity distributions, and perform extensive quantitative benchmarking against existing beam-blockage correction methods.

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